Cluster Robust Variance Estimation

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Table of contents

1 Introduction

In costructing the confidence intervals of the OLS estimator $\hat{\beta}_{\text{ols}}$, it is the asymptotic distribution of

$$\frac{1}{\sqrt{N}} \sum_{i=1}^{N} X_i u_i$$

that determines the estimator's asymptotic distribution. Assuming that $E[X_i u_i] = 0$, the Lindern-berg-Levy CLT apply:

$$\frac{1}{\sqrt{N}} \sum_{i=1}^{N} X_i u_i \xrightarrow{d} \mathcal{N}\left(0, \frac{1}{N} \sum_{i} \text{Var } (X_i u_i)\right)$$

But this hold true only if the individual observations are all independent, in fact summing 2 random variables X_1 and X_2 we get another random variable (call it Z) with expected value equal to $E[X_1] + E[X_2]$ and variance equal to $\text{Var } (X_1) + \text{Var } (X_2) + 2 \cdot \text{cov } (X_1, X_2)$. Summing N random variables $(X_1, ..., X_n)$, assume $E[X_i = 0]$, we get a new random variable $Z_N = \sum_{i=1}^{N} X_i$:

$$Z_N \sim \left(0, \sum_{i=1}^{N} \text{Var } (X_i) + 2 \sum_{i \neq}^{N} \sum_{j}^{N} \text{cov } (X_i, X_j)\right)$$

if all the X_is are **independent** all the joint moments (and so also the convariancers) are equal to zero and, dividing by $\sqrt{N}$:

$$\frac{1}{\sqrt{N}} (Z_N) \sim \left(0, \frac{1}{N} \sum_{i=1}^{N} \text{Var } (X_i)\right)$$

So the variance of Z_N scaled by $\frac{1}{\sqrt{N}}$ is the average variance of all the X_is. But here comes the problem, if our data are clustered we do not have that all the observation are independent, but rather we have that the observation of all the individuals inside a cluster are correlated, so we would have that

$$\frac{1}{\sqrt{N}} \sum_{i=1}^{N} X_i u_i \xrightarrow{d} \mathcal{N}\left(0, \frac{1}{N} \sum_{i=1}^{N} \text{Var }(X_i u_i) + \frac{2}{N} \sum_{i \neq}^{N} \sum_{j}^{N} \text{cov }(X_i u_i, X_j u_j)\right)$$

The White variance estimator $\xrightarrow{d} \frac{1}{N} \sum_{i=1}^{N} \text{Var }(X_i u_i)$, so it would not consistently estimate the asymptotic variance.

1.1 What is a cluster?

But let's take a step back and ask ourselves what does it mean that our data are clustered?
[1] define a cluster as a subgroup in the population such that individuals inside it has errors that are correlated. According to [2] what matters is the correlation of $X_i u_i$. This difference arises only due to the fact that in [1] regressors are considered non stochastic.
In essence, the defining feature of a cluster is that individuals within it exhibit some form of unobserved correlation.

> ⚠ Qualche esempio
>
> Qua devo aggiungere gli esempi dalla letteratura tipo:
> - B. R. Moulton [3]
> - M. Bertrand, E. Duflo, and S. Mullainathan [4]

Let's say that in our population there are N individuals from which we observe a trait X, so that we have N random variables $X_{ig} = (X_{11}, ..., X_{Ng1}, X_{12}, ..., X_{Ng2}, ..., X_{NgG})$, where i index an individual inside a cluster and g is the cluster. We can think of them as being drawn from the same distribution (so that they would be identically distributed), let's assume that this distribution has mean 0.
But they are not independent if they share the same cluster. This means that their joint distribution is not just the product of the marginals and their covariance may not be 0, so the convolution of two of them in the same cluster is:

$$X_{11} + X_{12} \sim (0, \text{Var }(X_1) + \text{Var }(X_2) + 2 \text{ Cov }(X_1, X_2))$$

Saying that observations within a cluster are correlated means that the values of one individual carries information about the values of another one in the same cluster. More concretely, if two individuals i and j belong to the same cluster g, then observing that one of them takes a value above the overall (population-wide) mean of X increases the likelihood that the other will also be above the mean:

$$P\left(X_i > E[X] \,|\, X_j > E[X] \text{ and } i, j \in g \right) > P\left(X_i > E[X] \right).$$

In other words, the event that X_j exceeds the population expectation $E[X]$ raises the conditional probability that X_i does the same when both belong to the same cluster. This would be the same for values below the mean:

$$P\left(X_i < E[X] \mid X_j < E[X] \text{ and } i, j \in g\right) > P\left(X_i < E[X]\right).$$

Outcomes within the same cluster are positively associated — they tend to move in the same direction relative to the overall mean.

1.2 Cluster-Specific Distribution

We can define $X_{ig} = \{X_i : \text{individual } i \in \text{ cluster } g\}$. X_{ig} can be thought of as a different random variable than X_i, with its own distribution. It is like zooming in a cluster, the probability of a certain value occurring would be different than the probability of the same value occurring in the whole population. It has its own expectation:

$$\mathbb{E}(X_{ig}) = E(X_i \mid X_i \in g) \equiv \mu_g$$

And, due to the law of total expectation:

$$\mathbb{E}(\mu_g) = \mathbb{E}[\mathbb{E}(X_i \mid X_i \in g)] = E[X_i]$$

It also has its own variance, which is the conditional variance of X_i, given that $X_i \in g$ Call G the event $X_i \in g$

$$\sigma_g^2 \equiv \mathbb{E}\left[X_{ig} - \mathbb{E}(X_{ig})\right]^2$$

$$= \mathbb{E}\left[(X_i - \mathbb{E}(X_i|G))^2 \middle| G\right]$$

$$\text{Var }(X_i \mid G) = \mathbb{E}[X_i^2 \mid G] - \left(\mathbb{E}[X_i \mid G]\right)^2$$

Due to the law of total variance:

$$\text{Var }(X_i) = \mathbb{E}[\text{Var }(X_i \mid G)] + \text{Var }(\mathbb{E}[X_i \mid G]).$$

2 Het

There are heterogeneous effect across clusters, this means that the true parameter β is different in every cluster.

$$y_{ig} = \alpha + \beta_g X_{ig} + u_{ig} \tag{1}$$

assume u_{ig} has a cluster component and an individual-and-cluster component:

$$u_{ig} = c_g + v_{ig}$$

Where $v_{ig} \sim (0, \sigma_v^2)$, $c_g \sim (0, \sigma_c^2)$ but of course c_g varies only at the cluster level.
In this way the expected value of the error given that individual i is in cluster g is:

$$\mathbb{E}_{\mathbb{g}}(u_{ig}) = c_g$$

But as long as we have an intercept we can assume w.l.o.g that $\mathbb{E}\left[X_{ig} u_{ig}\right] = 0$. If I estimate $\hat{\beta}_{\text{ols}}$ inside cluster g, I get:

$$\left(\widehat{\beta}_g - \beta_g\right) = \left(\sum_{i \in g} X_i^2\right)^{-1} \left(\sum_{i \in g} X_i u_i\right)$$

And:

$$\hat{\beta}_{\text{g}} \overset{a}{\sim} \mathcal{N}\left(\beta_g, \frac{1}{N_g}\left(\sum_{i \in g} X_i^2\right)^{-1} B \left(\sum_{i \in g} X_i^2\right)^{-1}\right)$$

where B_g is:

$$B_g = \text{Var}\ \left(X_{ig} u_{ig}\right)$$

If I estimate β across G clusters I get a vector of G $\hat{\beta}_{\text{g}}$, and for each of these I can consistently estimate an asymptotic variance using White variance estimator since observation inside a cluster are i.i.d. Since a linear combination of normal is itself normal I can average out those variances and get an estimate of the Avar of $\hat{\beta}_{\text{ols}}$.

To test this let us simulate a dataset, first we draw a random vector of betas from a normal distribution with mean 2 and variance 4.

```
include("../src/ClusterSimulation.jl")
G = 50 #Number of clusters
Ng = 1000 # Number of individuals per cluster

β = rand(Normal(2,4),G)
```

```
50-element Vector{Float64}:
 -1.096752120944871
  5.641715464047893
  7.293205581265417
  5.786338681759574
 -2.7752345760788746
  3.187101203777012
  5.751764096211713
  4.778186039882959
  6.334790803231912
  2.3813106587340003
  7.954176351491988
 -2.9966903763889965
  4.5035016431358095
  ⋮
  0.18250621946541012
  3.4992694803095645
 -5.52902477576072
  2.2813448861387804
  4.299236706970355
```

```
  2.455769337038656
  9.381534494937384
  0.20696710727861833
  3.2741482960701713
  6.058439095234678
  2.2331000669773595
 13.447020898459298
```

Then we use each of these betas to simulate a cluster according to the model in Equation 1

```
pop = hePopulation(G,β,randn(G))
```

```
hePopulation(50, [-1.096752120944871, 5.641715464047893, 7.293205581265417,
5.786338681759574, -2.7752345760788746, 3.187101203777012, 5.751764096211713,
4.778186039882959, 6.334790803231912, 2.3813106587340003  …
-5.52902477576072, 2.2813448861387804, 4.299236706970355, 2.455769337038656,
9.381534494937384, 0.20696710727861833, 3.2741482960701713, 6.058439095234678,
2.2331000669773595, 13.447020898459298], [0.008502495902945047,
-0.9008907694717521, -0.577908381476869, 2.171627216613601,
-0.047890982157932305, -0.6445952853525871, 0.1642978934984976,
-1.323786401204805, -0.2019430693691375, 0.97408479714506  …
-0.2702081709979929, -2.0183631777765036, 0.2759079618644971,
0.8900673891393104, -1.4593989235503224, 0.44627642124731076,
-0.08855343207946456, 1.8044462801370542, 0.9015040611232071,
-1.4449524515646368])
```

This line creates a heterogeneous-effects population.
- **`G`** is the number of clusters.
- **`β`** is the vector of betas.
- **`randn(G)`** produces the vector $c_g \sim \mathcal{N}(0, 1)$.
The constructor returns a population struct that stores the cluster-level parameters used later to simulate data.

```
sam = sample(pop, Ng)
```

```
Cluster sample struct
```

This line generates a sample from the population.
- **`Ng`** is the number of observations per cluster.
The function draws data for all `G` clusters, each containing `Ng` observations, and returns a sample struct built according to the population's parameters.
The sample struct has a field `allclusters` that is an array of cluster structs, so we can define a function that compute $\hat{\beta}_{\mathrm{g}}$ in each cluster and then broadcast it to all clusters in our sample, getting G estimates.

```
betahat = permutedims(stack(bols.(sam.allclusters)))[:,2]
```

```
50-element Vector{Float64}:
 -1.116124572729701
  5.597978265298389
  7.344997638447929
  5.7858866585239355
 -2.776798215308975
  3.2072218592051778
  5.763944696192871
  4.760387292145575
  6.359145163284593
  2.3707553513453443
  7.959879815086764
 -2.9922429465565514
  4.504200540744616
  ⋮
  0.20612191960314596
  3.5400850762003375
 -5.536392244062465
  2.258082021159959
  4.291720307653179
  2.4822821563621633
  9.356222820370641
  0.21087555639137706
  3.302102384691868
  6.075903911424142
  2.2368050429896473
 13.45476527428469
```

And similarly define a function that compute the White Variance estimator and broadcast it to all clusters.

```
WhiteAvar.(sam.allclusters)
```

```
50-element Vector{Matrix{Float64}}:
 [0.0009963789176529018 2.9989249313604236e-5; 2.998924931360433e-5
0.0009993391599721155]
 [0.0010949385127685504 -1.5040577167364637e-5; -1.5040577167364622e-5
0.0010586185700855026]
 [0.0009476404301304026 -2.8426928013916853e-5; -2.842692801391685e-5
0.0007935903463968676]
 [0.0008796708796458699 1.967338000163768e-5; 1.9673380001637683e-5
0.0007153425222854338]
 [0.0010380701383350271 -6.882849591663562e-5; -6.882849591663565e-5
0.0009027713731049637]
```

```
 [0.001005644344769767 1.4099440218529478e-5; 1.4099440218529529e-5
0.0010089236968802632]
 [0.0009737007951793469 1.0714998724115744e-5; 1.0714998724115805e-5
0.0009638321639582678]
 [0.0009943255636611422 8.01150721085487e-6; 8.011507210854904e-6
0.001080809876386774]
 [0.0010104259225066827 2.637298912618618e-5; 2.637298912618618e-5
0.0011283953957557688]
 [0.000968556251255066 3.417044217762518e-5; 3.417044217762524e-5
0.0009385055882686098]
 [0.0010487701799792469 4.66318837944298e-5; 4.663188379442979e-5
0.0010296305057724616]
 [0.0009577257126446804 3.562539925868523e-5; 3.5625399258685275e-5
0.0009483083345707205]
 [0.0009551130959455509 -4.044484499687414e-5; -4.0444844996874116e-5
0.0008566663513228224]
 ⋮
 [0.0009794713684143407 7.127475996156915e-5; 7.127475996156912e-5
0.001089615914580078]
 [0.0009963114797882308 -1.4047441054650809e-6; -1.4047441054650572e-6
0.0009645288243068825]
 [0.0009671177511437932 3.2136259832100245e-5; 3.2136259832100224e-5
0.0008708492994685505]
 [0.0009954000689233683 -6.086095968290202e-5; -6.086095968290201e-5
0.0009860538599953913]
 [0.0010889287164942546 -0.00015551917581398822; -0.00015551917581398835
0.0011664525057467432]
 [0.001119639359593166 -9.693671325827424e-5; -9.693671325827419e-5
0.0013332094211182491]
 [0.0008700302905248731 -4.91204191388023e-6; -4.912041913880251e-6
0.0008830610309797048]
 [0.0010407030959826923 3.3177611060971325e-5; 3.317761106097135e-5
0.0010282516736749888]
 [0.0010080129891032708 8.702023798476676e-5; 8.702023798476676e-5
0.001004595879157276]
 [0.0009265162669604297 -2.4501752642864287e-5; -2.4501752642864266e-5
0.0007741399311577665]
 [0.0010325147650796826 0.00011274886036282097; 0.00011274886036282108
0.0010540524547565967]
 [0.0009698063799777551 -1.4667199754736391e-5; -1.466719975473635e-5
0.0009992869171058124]
```

We are interested in the average across clusters of β_g, the average partial effect. Note that since we have a finite amount of clusters, each of them with the same size, the APE would be:

$$\beta_{\text{APE}}\frac{1}{G}\sum_{g}\beta_g$$

Note that doing $\frac{1}{G} \sum_g \hat{\beta}_{\mathrm{g}}$ is the same as computing $\hat{\beta}_{\mathrm{ols}}$ in the whole sample. We define a function to do so:

```
function bols(s::clSample)
    X = vcat(getallX(s)...)
    Y = vcat(getallY(s)...)
    betahat = inv(X' * X) * (X'Y)
end
```

```
bols (generic function with 2 methods)
```

3 Consistency as $N_g \xrightarrow{p} \infty$

We want to test wether our estimate of the average partial effect gets better if the number of people that we draw in each cluster increases. In order to do so we ran a series of montecarlo simulation increasing N_g.

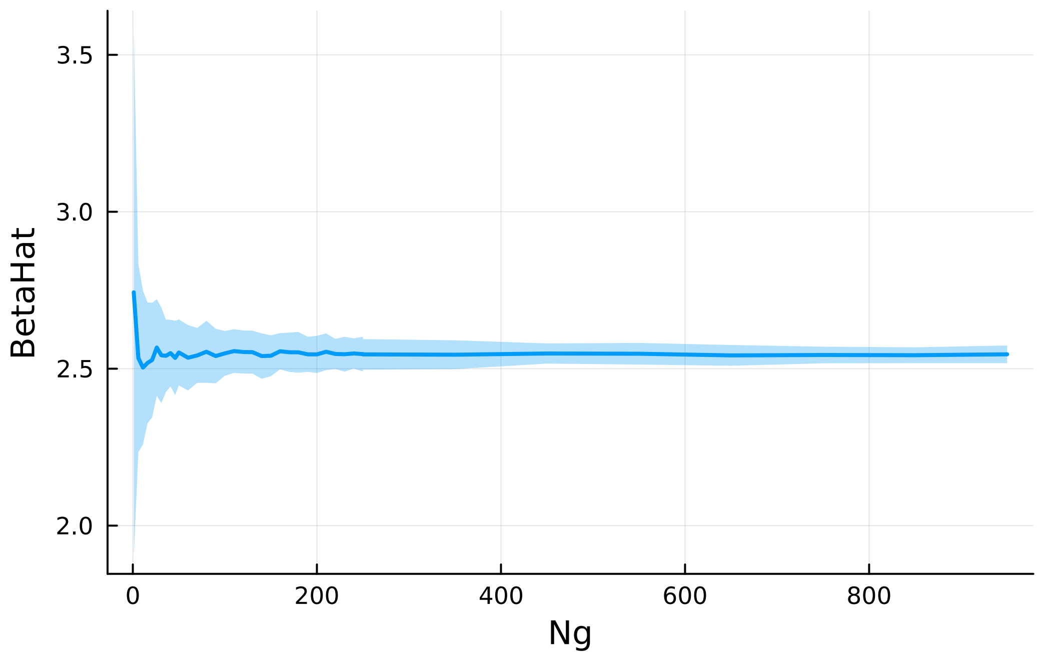

Figure 1: Consistency of betahat as Ng

What we see in Figure 1 is the result of the montecarlo simulation, the line is the average estimates for each N_g, while the shaded area is the standard deviation, we can see clearly see from the graph that as the number of observations inside a cluster grows $\hat{\beta}_{\mathrm{ols}} \xrightarrow{p} \beta_{\mathrm{APE}}$.

Bibliography

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